Algebra Tutorial 1: Binary Operations, Groups

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1. Binary Operations

A binary operation on S is a map

$$* : S \times S \to S.$$

1. Which of the following are binary operations?
   (a) $a * b = a + b$ on $\mathbb{Z}$;
   (b) $a * b = a - b$ on ;
   (c) $a * b = ab$ on $\mathbb{Z}$;
   (d) $a * b = a/b$ on $\mathbb{Q}$.
2. Determine which of the following operations are associative and which are commutative:

$$a * b = a + b, \qquad a * b = a - b, \qquad a * b = ab.$$

3. Give an example of an associative but non-commutative operation.

2. Neutral Elements and Inverses

e is a neutral element if

$$e * a = a * e = a.$$

b is an inverse of a if

$$a * b = b * a = e.$$

4. Find the neutral element, if it exists:
   (a) $a * b = a + b$ on $\mathbb{Z}$;
   (b) $a * b = a + b - 3$ on $\mathbb{Z}$;
   (c) $a * b = ab$ on $\mathbb{Q}$;
   (d) $a * b = a + b + ab$ on $\mathbb{R}$.
5. Find the inverse of a in each case:
   (a) $(\mathbb{Z}, +)$;
   (b) $(\mathbb{Q} \setminus \{0\}, \cdot)$;
   (c) $a * b = a + b - 2$ on $\mathbb{R}$.
6. Prove that a neutral element is unique.
7. Prove that inverses are unique whenever the operation is associative.

3. Groups

A group $(G, *)$ satisfies:

$$\begin{aligned} &a * b \in G, \\ &(a * b) * c = a * (b * c), \\ &\exists e \in G : \quad ea = ae = a, \\ &\forall a \in G \ \exists a^{-1} \in G : \quad aa^{-1} = a^{-1}a = e. \end{aligned}$$

A group is *abelian* if $ab = ba$ for all $a, b \in G$.

8. Determine which are groups, and state whether they are abelian:
   (a) $(\mathbb{Z}, +)$;
   (b) $(\mathbb{Z}, \cdot)$;
   (c) $(\mathbb{Q}, +)$;
   (d) $(\mathbb{Q} \setminus \{0\}, \cdot)$;
   (e) $(\mathbb{R}, +)$;
   (f) $(\mathbb{R} \setminus \{0\}, \cdot)$.
9. Let
$$G = \{1, -1, i, -i\}$$
under multiplication. Show that G is an abelian group and find all inverses.
10. Let G be a group. Prove
$$(ab)^{-1} = b^{-1}a^{-1}.$$
11. Prove the cancellation law:
$$ab = ac \implies b = c.$$
12. Prove that
$$a^2 = e \implies a^{-1} = a.$$

4. Finite Groups and Cayley Tables

A finite group is a group with finitely many elements. Its order is $|G|$.

13. Complete the Cayley table of $\mathbb{Z}/3\mathbb{Z}$:

$+$	0	1	2
0			
1			
2			

14. Complete the Cayley table of $\mathbb{Z}/4\mathbb{Z}$:

$+$	0	1	2	3
0				
1				
2				
3				

15. Find the inverse of every element of $\mathbb{Z}/6\mathbb{Z}$ under addition.
16. Compute in $\mathbb{Z}/7\mathbb{Z}$:
$$[4] + [6], \qquad [5] + [5], \qquad [3] + [3] + [3].$$
17. Solve in $\mathbb{Z}/8\mathbb{Z}$:
$$[x] + [3] = [6].$$
18. Show that $\mathbb{Z}/n\mathbb{Z}$ is an abelian group under addition modulo n.

5. Groups of Units

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{[a] \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}.$$

19. Determine $(\mathbb{Z}/10\mathbb{Z})^\times$ and find the inverse of every element.
20. Determine $(\mathbb{Z}/8\mathbb{Z})^\times$ and construct its Cayley table.
21. Find the inverse of 5 in $(\mathbb{Z}/12\mathbb{Z})^\times$.
22. Explain why $[2] \notin (\mathbb{Z}/8\mathbb{Z})^\times$.

6. Permutation Groups

S_n is the group of all permutations of $\{1, \ldots, n\}$ under composition. In particular,

$$|S_n| = n!.$$

23. List all elements of S_3 in cycle notation.
24. Compute
$$(123)(12), \qquad (12)(123).$$
25. Find
$$(123)^{-1}.$$
26. Give two elements of S_3 which do not commute.
27. Explain why S_3 is a finite non-abelian group.

7. More Finite Groups

28. Consider
$$V_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$
List its elements and construct its Cayley table.
29. Compare $\mathbb{Z}/4\mathbb{Z}$ and V_4. They both have order 4. Give a property showing that they are different groups.
30. The symmetries of a regular n-gon form the dihedral group D_n. How many elements does D_n have?
31. Construct a group of order 4 which is not $\mathbb{Z}/4\mathbb{Z}$.
32. Find all elements of $\mathbb{Z}/12\mathbb{Z}$ having additive order 3.

8. Challenge

33. Let G be a group and suppose $ab = e$. Prove that $ba = e$.
34. Let G be an abelian group. Prove
$$(ab)^n = a^n b^n$$
for every $n \geq 1$.
35. Find all $x \in (\mathbb{Z}/15\mathbb{Z})^\times$ satisfying
$$x^2 = [1].$$